# MANTEJ SOKHI

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## EDUCATION

**M.Sc. in Applied Mathematics**, Simon Fraser University — Expected 2027
**B.Sc. Honours in Applications of Mathematics**, Kwantlen University — 2018 – 2023

## EXPERIENCE

**Graduate Researcher** — Sept 2024 - Present
Center for Experimental and Constructive Mathematics — Burnaby, BC

- Designed and built C++ and C libraries for fast finite field polynomial and integer arithmetic and integrated them into Maple through its foreign function interface.
- Implemented rational reconstruction and Newton interpolation in C++, beating Maple's built-ins (Ratrecon and Interp) by **209**× with a 32 bit prime and reducing black box probes by **96**×.
- Implemented Maple's modular Linear solve routine in C++ with batched FFI calls, reducing per call latency from **1075** micro seconds to **6.9** micro seconds (**900**×). Extended the solver to handle 64 bit primes, unsupported by Maple, which is limited to 32 bit hardware integers.
- Designed heuristic and deterministic fault-tolerant variants of the reconstruction algorithms, developing the supporting subresultant theory and failure probability analysis to verify correctness over Galois fields.
- Benchmarked the pipeline on a family of parametric linear systems, outperforming Maple on every case and co-authored a peer reviewed paper (ICMS 2026) on the implementation.

**Software Engineer** — Feb 2024 - May 2024
Inverted AI — Vancouver, BC

- Built Python cleaning and transformation pipelines that standardized large autonomous driving datasets (100 videos of ∼12 min per country) into consistent, simulation ready inputs for model training.
- Diagnosed recurring CVAT labeling defects by tracing error patterns to their source data and correcting the affected records, stopping **45**% of corrupt annotations from propagating into simulation and training.
- Built Python tools that overlaid annotations onto JOSM base maps, letting the team visually review their drawings and catch geometry errors before validation.

**Research Assistant** — March 2023 - December 2023
Kwantlen University — Surrey, BC

- Built Python simulations with NetworkX, NumPy and Pandas to test discrete pursuit evasion strategies across combinatorial graph families, checking that computed damage numbers stayed within derived theoretical bounds.
- Developed and proved the underlying theory and asymptotic bounds, contributing results to a peer reviewed publication.

## PROJECTS

**Vector Search Engine.** Built exact kNN and HNSW search in C++ with Python bindings, multithreaded queries, and index persistence. Achieved **98.2% Recall@10 at 1,850 QPS** on SIFT1M, an **8.4× speedup** over exact search.

**Distributed Key-Value Store.** Built a concurrent Go TCP store with write-ahead logging, crash recovery, LSM storage, Bloom filters and Raft replication. Sustained **105K ops/s** under a 70/30 workload and **19K replicated writes/s at 14 ms p99** across three nodes with zero acknowledged write loss.

**Navier-Stokes Solver.** Implemented a C++ 2D lid driven cavity solver on a staggered MAC grid with upwind advection and Jacobi/SOR pressure Poisson solvers. Added residual, divergence and CPU cycle diagnostics.

**Polynomial GCD Verification.** Formally verified finite field polynomial arithmetic and Euclidean GCD in Dafny with preconditions, loop invariants, termination metrics and proof lemmas. Compiled and tested the verified code across multiple target languages.

## TECHNICAL SKILLS

**Languages** — C++, C, Golang, Maple, Matlab, R, Python, Julia, SQL.
**Tools & Infrastructure** — Git, Bash, Linux, CMake, Docker, CI/CD Pipelines, Observability (Logging, Metrics, Tracing).